

Prevention of PSE^{*}

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Abstract. Photosensitive epilepsy is a disease that can cause discomfort when exposed by visual triggers. Some visuals are more triggering than others; Eg. the frequency of a display, opposite colour contrast, and static patterns. Conducted research proves that these triggers can be avoided by preventive measures such as reducing the brightness of a screen, adding blocks to possible triggers, and warning the user beforehand.

Keywords: Photosensitive Epilepsy · Prevention · Program.

1 Introduction

Photosensitive epilepsy (PSE) is a form of epilepsy which can cause seizures or discomfort for the ones suffering from it. Around 3 to 5 percent of epileptic people suffer from photosensitive epilepsy. Children and adolescent people tend to surf a lot on the internet. These who suffer from photosensitive epilepsy could unexpectedly run into triggering content. We are going to solve this by creating a program using AI that detects and blocks these triggering stimuli in advance. We will answer the following questions: What is the problem? (Domain and task) What has been done? (Literature and Knowledge) How can the problem be solved? (Computational solution) Is the solution valid? (Evaluation plan)

2 Domain and Task

2.1 Domain

Photosensitive epilepsy (PSE) is a form of epilepsy which can cause seizures or discomfort for the ones suffering from it. Several visual stimuli we encounter in daily life can trigger such reactions. Think about watching the television, walking on the streets and seeing a flickering billboard, visiting a flashy website while shopping online, or playing a video game containing flickering lights. These visual images or lights can be a huge deal for people who suffer from PSE. For our domain we focus on people who suffer from photosensitive epilepsy (PSE) that

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surf on the internet. Since most games containing flickering lights already have a warning for people that suffer from photosensitive epilepsy, we will not include video games in our research, and only focus on standard websites one visits while just browsing instead. Those who suffer from this disease are mostly children and adolescents, and the increase in young people on the internet is rapidly growing. These people are sensitive to flickering lights and/or certain types of patterns. Exposure to these triggers, both in real life and on the internet, can cause seizures and/or discomfort in people suffering from PSE.

2.2 Problem

As previously mentioned, children and adolescent people tend to surf a lot on the internet. There is a rapid increase in websites that are more advanced and flashy looking, containing flickering lights and cool widgets that make their site look better. While browsing, these who suffer from photosensitive epilepsy could unexpectedly run into these stimuli, which can be a trigger for them. They do not know such a site contains flickering lights and/or certain patterns that might trigger their seizure/discomfort, hence creating a problem for them to surf the internet without concern for their health. We are going to solve this by creating a program using AI that detects and blocks these triggering stimuli in advance.

2.3 Input

Photosensitive epilepsy (PSE) is a form of epilepsy in which seizures are triggered by visual stimuli that form patterns in time or space, such as flashing lights; bold, regular patterns; or regular moving patterns. Stimuli that cover one's entire visual field are more triggering than stimuli that cover only parts of one's visual field. Even though it is less triggering if only small parts of one's visual field is covered, it is still a problem for lots of people with epilepsy that do not get a warning before entering a website. Thankfully, we could use AI technology to minimize discomfort and/or seizures in people suffering from PSE. The input of our program/app will be the flickering lights, contrast in colors, stripe patterns, dot patterns and rapidly changing and alternating images that will let the program detect whether it can be a trigger for people who have epilepsy. If we show the program lots of visual stimuli, it should be able to recognize these triggers on the internet. We should make the program learn what it does not have to recognize as well, in that way the machine can learn from its mistakes. Also, we will interview an expert that knows more about photosensitive epilepsy and someone who suffers from it their selves, so we can know what people suffering from epilepsy should be wary of.

2.4 Output

The output will darken your screen and send a pop-up, warning the user that the site they are trying to enter might contain stimuli that might cause discomfort for

people with photosensitive epilepsy. It will explain what the site might contain: bright, flashing lights and/or imagery that may cause the discomfort or even a seizure. This will then give the user the option to either leave the site, or continue despite the warning. When the user consciously chooses to enter the site, certain components on the website that might contain triggers, will be blocked with a panel, telling the user what is beneath the panel that might be a trigger. The user can still choose to view the blocked content by clicking the panel away, since our program can still make mistakes; maybe the blocked content is not a trigger for the user at all. Furthermore, the program will also present a warning to the user whenever he or she has been on their computer for too long, as this could also trigger a seizure. This way, lots of seizures and/or discomfort can be prevented while leisurely browsing on the internet.

2.5 Knowledge That is needed

We will have to know what things can trigger a seizure and/or discomfort for someone with photosensitive epilepsy and what will help avoiding these triggers using AI. To obtain this knowledge, we will interview two experts: one expert in the field of epilepsy and someone who suffers from photosensitive epilepsy themselves. We will also dig deeper into epilepsy using some literature (books/researches) about photosensitive epilepsy. Later on, we will need to have knowledge of Computer Science (to write such a program), Artificial Intelligence (to make our program “understand” the patterns in which the seizure occurs) and some psychological/mental information to tell us how the seizure is triggered, but we will dig into that in a later assignment.

3 Literature and Knowledge Acquisition

3.1 Planning of the knowledge acquisition

We will be mostly collecting information by looking for several articles and research projects from Google Scholar. The workload will be divided and we will be scanning through the online articles and writing short summaries about them to derive a final summary. Another way of obtaining information about photosensitive epilepsy is through interviews. We have planned an interview with someone who suffers from photosensitive epilepsy, and an expert from an epilepsy foundation called ‘Epilepsilijn’. This way, we can learn about epilepsy through different perspectives and derive a solution for people with epilepsy browsing the internet.

3.2 Summary of the literature

Photosensitive epilepsy (PSE) is a form of epilepsy which can cause seizures or discomfort for the ones suffering from it. Several visual stimuli we encounter in daily life can trigger such reactions. Think about watching the television, walking on the streets and seeing a flickering billboard, visiting a flashy website

while shopping online, or playing a video game containing flickering lights.[1, 2, 8] These visual images or lights can be a huge deal for people who suffer from PSE.

One of the articles we found was a research into what is most triggering for photosensitive epilepsy. Certain patterns cause more triggers than others. In addition, certain colors may cause more triggers. An example would be that the colour red is more triggering than white.[3] Also, opposite color contrasts are also triggering. One would say static patterns are one of the patterns most vulnerable to epileptic people, like the static from a television screen; the image of a static pattern is continuously in motion.[10, 4] Brightness and contrast also play a big role when talking about epilepsy triggers.[9] This information is important to keep in mind when making our AI program. There has already been a lot of research on how to live with photosensitive epilepsy. One article wrote that a lot of patients wear sunglasses when going out.

Another source about the treatment of PE and how to avoid a seizure wrote that drug treatment is not always the most useful way to avoid a seizure.[7] It depends on the environment of the patient and the frequency and severity of his or her epilepsy. Medicines such as VPA CLB are deemed to be effective to prevent seizures. Anti epileptic drugs (AEDs) are not preferred as the first-line therapy.[5] However, stimuli avoiding techniques such as video hazard blocks, special TV screens, and glasses should be enough to prevent a seizure.[6]

We want to write an AI program that reduces the brightness, adds blocks covering the possible trigger, or makes colors less bright when triggering events are happening on the screen.

3.3 Impact of the Interviews

The epilepsy foundation provided information about epilepsy and its different forms of seizures. The different preventive measures they gave us, aided us with choosing the right solution for our project. Because of our call, we now have a better understanding of what people with epilepsy go through on a day to day basis, and how to implement our solution with little to no interference in their life. They gave us examples of common triggers which helped us envision how it must feel like to live with photosensitive epilepsy. Additionally, the epilepsy fund gave us different tips on who we could potentially interview next. Although we might still make some changes to the given solution. We believe that talking to the epilepsy fund put us in the right direction and we genuinely hope to make a difference in their life as the epilepsy fund is doing.

The interview with the expert who suffers from photosensitive epilepsy himself was also very helpful to us. Of course, there is no one to better understand the impact of a trigger than someone who is actually bothered by those triggers. For example, it was new to us that a trigger does not necessarily lead to a seizure but can also just make him feel not well. In addition, he gave us some information about the medication that we had not found on the internet yet, like the side effect of gaining weight. Moreover, he told us that he does not think there are enough warnings for triggers on the internet. This is, of course, very

unpleasant for him, nevertheless it was good to hear that our AI is really needed and that it does not exist already.

3.4 Reflection on the Knowledge Acquisition session

This session, given by Stefan Schlobach, taught us the best strategy in finding literature. Not only in finding the appropriate literature, but also to judge the literature on their significance and to connect the information with each other. The KA session also gave us some insight on how to conduct the interviews, ask the relevant questions, how to approach the interviewees and overall how to have a structured interview. Furthermore, the session also warned us about common pitfalls and how to prevent them from happening during the interviews. For example, we avoided long sessions and we transcribed the interviews as soon as possible. Overall, the lecture had a great impact on us and on how we did our research, and made it easier for us to find our sources.

4 Computational Solution

4.1 What data is being collected?

Data about triggering visions is needed. In order to make an AI that can recognize triggers we need to have sufficient knowledge about these triggers. Examples of triggering visions are static patterns, color contrasts, flickering lights and certain colours.

In addition to this we need data to define the triggers. For example, ‘How much should a light flicker to be considered triggering?’, or ‘Which color contrasts are more triggering?’. Based on the definition of the different triggers we will find examples of these triggers and show these to the AI.

Lastly, we should collect data about the different machine learning techniques to learn the limitations of what our AI can and cannot detect.

With the knowledge collected about machine learning and triggers, the AI should be able to recognize new triggers by itself as we put more examples in the input. With all this knowledge together, our AI should be fully functioning.

4.2 Which algorithms are used to analyse the data?

To analyse the data, a machine learning algorithm will be used in combination with an image recognition algorithm to detect triggering patterns such as flickers, contrasts, patterns, etc. The image recognition algorithm should be able to detect different triggers based on the examples we put in our input. The machine learning algorithm will then, based on these examples, learn how a trigger would look like and thus find them by themselves.

See Appendix B for the Flowchart

Explanation of the flowchart The input of the flowchart is the screen of the user. The flowchart contains the triggers that the user might encounter when using the internet. With every trigger there is a parallelogram with the solution our AI will give. Our AI is a program you can download when you are sensitive to certain triggers or suffering from photosensitive epilepsy. It is an optional program that you can turn on and off when surfing on the internet. Also, when a warning appears you still get the option to enter the website or continue what you were doing.

4.3 Which results are being computed, and how?

The AI will scan the internet page that you are trying to visit or the video/movie that you want to watch before showing it to you. If it finds anything that looks like the examples that we have shown the AI, it will be able to detect this trigger and warn the user. Because of machine learning, the AI knows what it should recognize as a trigger and what can be shown to the user. The triggers have been categorized, so that each trigger will have a suitable solution to visit the website anyway. For example, if there are color contrasts, the screen will be darkened. If the AI did not find anything like a trigger on the site or in the video, it will just be shown to the user as it normally would. It will not show a message saying “No trigger found” or anything like that, since it would be rather annoying to be shown a notification every time you click on a website or video.

4.4 How will this be presented to the user?

As we have shown in our flowchart, there will be different outputs for their specific inputs. When the user encounters certain patterns, like dots and stripes, the patterns will be turned into a solid area. When our AI program detects flickering lights and frequently alternating images, these areas will be covered with blocks displaying a warning to the user. Another feature of our AI program will be a darkened screen when huge colour contrasts are detected. For all these types of safety measures, the program gives the user the freedom of choice to still view the possible triggers; they could remove the warning blocks, put a stop to the dimming of the screen, and undo the changed patterns.

5 Evaluation Plan

To test our AI program, we could first check ourselves and see if the program does its job. Afterwards, we could ask several people that suffer from photosensitive epilepsy to download our program and use it on several websites that contain flickering lights. This way, we could see if our program is reacting fast enough to block any uncomfortable images. We could make up a questionnaire to ask users their experience with our program; was it easy to use? Was it effective? Did you feel safer while using it? Is there anything missing?

5.1 Internal validation

Our internal validation will be separated in learning and testing the AI program. 60 percent of our data (pictures of triggers, etc.) will be given to the machine so that it can recognize these triggers for our system. This is the learning part. We will save 40 percent of our data to test the AI program and see whether the AI program is doing what it is supposed to do. So, if we show this 40 percent to the AI program, it should know whether this is a trigger or not. If it does not work well enough yet, we can always use a part of this 40 percent to teach the machine more about what triggers to recognize.

5.2 External validation

We are going to need volunteers to test our AI program. These volunteers need to suffer from photosensitive epilepsy. We only accept test applicants that have a light form of photosensitive epilepsy, since we would not want anyone to immediately experience a seizure when something goes wrong with our AI program. We are going to expose these applicants to websites that contain triggers and test whether our alternatives will have an effect on them. All different types of triggers will have to be tested, including the solutions the AI program comes up to cover up those triggers. The applicants will need to fill out a questionnaire asking which websites are most harmful to them. The websites are selected based on the results of the questionnaire. The AI program will be tested in a quiet room where no other triggers for epilepsy can occur. This way, we know that a possible seizure can only be caused by triggers on the computer screen. After the experiment, we will conduct a questionnaire on user satisfaction of our program. By doing this, we will know what is good about our program and what needs some adjustments. The volunteers will receive an amount of money for their participation.

5.3 Hypothesis

We expect that the test for the external validation will provide positive results, meaning that the websites will have no effect on the participants. If the websites do have an effect on the participants, we will amplify the blocking of the photosensitive triggers until we reach the point that no participants will be affected. That way we can see when the AI program will be actually useful.

5.4 Power analysis

For the power analysis, we need to know our study group design, primary endpoint and some statistical parameters. Those statistical parameters are the anticipated incidence (known population vs. study group) and the type I/II error rate (alpha and power).

Alpha = the probability of a type-I-error, determining that there is a difference between two groups when such difference does not actually exist (false

positive rate). Most medical literature uses a rate of 0.05, so that is the rate we will use as well.

Power = the ability to detect a difference between two groups when a difference actually exists. Power is $1 - \beta$, where β is the risk of a type-II-error (false negative rate). Most medical literature uses a rate of 80-90 percent power (β of 0.1 - 0.2), we will be using a rate of 80 percent.

For our study group design, we have chosen “one study group vs. population”, as the people who suffer from photosensitive epilepsy are only a small group in the population. For the primary endpoint, we have chosen “dichotomous”, because our study group will either be triggered or not, there is no average.

Explanation of power analysis There are 120.000 people who suffer from epilepsy in the Netherlands. 3 - 5 percent of the epileptic people suffer from photosensitive epilepsy. $(120,000 / 17,000,000) * 100 \text{ percent} = 0.71 \text{ percent}$ of the dutch population suffers from epilepsy. $0.04 * 120.000 = 4800$ people who suffer from photosensitive in the Netherlands. $(4,800 / 17,000,000) * 100 \text{ percent} = 0.028 \text{ percent}$ of the dutch population suffers from photosensitive epilepsy. So: known population = 0.71 percent and study group = 0.028 percent

See Appendix C for Power Analysis

5.5 Ethical considerations

Even though the ethical acceptability is still a problem, the potential benefits (for everyone with photosensitive epilepsy) will outweigh the risks. By conducting this kind of experiment, we can make sure the internet is safe for everyone, even for people with epilepsy. By using blocks and other features, we can avoid any uncomfortable triggers. This way, nobody has to be scared to encounter any visual triggers for photosensitive epilepsy anymore.

5.6 Risks to participants

Even though we have walked through every step in our program and are sure the program is working as expected, we need to make clear what the risks are to the people who want to try our system. We make a budget available for the users that want to test the AI program. We only want to ask people to test our AI program with a light form of photosensitive epilepsy. This way we ensure that no major health issues will occur because the safety of our volunteers will be our highest priority. The risks of the system are the risks that you as an epilepsy patient are familiar with.

5.7 Overall evaluation

We have taken many factors in consideration while looking for the best possible AI program for photosensitive epilepsy patients on the internet. Through internal validation, we figure that our system will probably work perfectly fine with

the different data we can input to test our AI program. Through external validation, we will find new aspects to focus and improve on, and find the best user experience. Despite our small risks for people participating in our experiment, the results will increase the benefits for our public health; everyone needs to feel safe on the internet! We can conclude from the above mentioned considerations, that our AI program will tick every box in conducting a safe and comfortable online environment for people suffering from photosensitive epilepsy.

6 Conclusion

Combining all the information together, one should be able to make a fully functioning AI. We thought this course was solely focused on getting a better understanding of what AI was and how it should be utilized. However, we also learned the importance of doing good research and the different processes and considerations of creating and testing an AI. Although we did not have the privilege to build the AI, the research that was done to design the AI gave us insight of all the challenges an AI designer faces. This will definitely help us in our further academic career.

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A Interviews

A.1 Interview epilepsy helpline

How does the epilepsia fund help people with epilepsy? We provide educational information about epilepsy for people who are new to the topic. We also help book holiday trips for people with this disease and conduct scientific research. Our mission is to improve the quality of life for these people.

Is there a drug that could potentially reduce the chance of a seizure?

There is enough. But to trace it back to the subject of matter, photosensitive epilepsy, it is better to prevent than to treat. For example, there are certain measures they can take: - Stay at a 2-meter distance from your screen - Watch TV in a well-lit room - Take breaks between computer sessions - Wear sunglasses or a cap when outside - Reduce the brightness of your screen Just avoid anything that has a lot of contrast.

On your website we read that about 3-5 percent of people with epilepsy are photosensitive. How many cases of people with photosensitive epilepsy are there in the Netherlands / in the world? There are around 120.000 people with epilepsy in the world, around 4-5 thousand of these suffer from photosensitive epilepsy. I do not know the exact amount of people with photosensitive epilepsy in the world.

Are there any factors that increase the chance of a seizure for people with photosensitive epilepsy (think of stress, exam week, sadness, etc.)? Yes. Fatigue, stress, and over excitement are factors that play a noticeable role. They make people that suffer from photosensitive epilepsy more prone to having a seizure.

Is it true that digital triggers are as harmful as real life triggers? It depends on how these people deal with it. If they take active preventive measures, they will be less bothered by digital triggers. This is because they have more control over digital triggers. In real life, they have little to no control over the triggers. But in general, I would like to say that they are equally harmful.

Is there a difference in the intensity of attacks (is one attack more intense than the other)? Yes. Nor is it that everyone with epilepsy have those spastic movements that are displayed on tv. There are different forms of epileptic seizures. Some people do suffer from jerky movements while having a seizure, while others just stare into space.

Do you know anything about online trigger warnings, and do you think there should be more warnings online? Many console companies such as XBOX (Microsoft) have already integrated something like this in their games. I also know that some sites give out warnings whenever a trigger might appear. But I do not know where else. If I must deduct an answer from the questions asked to our helpline, it is useful to give more warnings on the internet.

A.2 Interview experience expert

How long have you been living with epilepsy? About 4 years ago I had my first seizure and since then I have had it regularly. In retrospect it turns out that I had epilepsy for some time, but that I have never noticed anything before.

How much does it affect your life? In the beginning, the epilepsy didn't have much of an impact, but it certainly did in the last two years! I cannot go to sleep late every night and wake up early the next morning, nights of at least 8 hours are really necessary for me. For example, getting my driver's license also causes a lot of problems, because I have to be seizure-free for a year before I can start lessons. Furthermore, I feel different when I am with friends at a party with a lot of flashes of light, I am already slightly triggered.

Have you ever experienced an attack? If so, what caused this attack? So far I have experienced epileptic seizures three times. In the first attack the cause was unclear, this happened on a normal day at school. The second attack was caused by not taking my medications consistently enough and sleeping very badly for two nights. The third attack was because I had drunk alcohol two nights in a row, I had not slept much and I had also forgotten my medication once in those two days.

What kind of triggers are you most sensitive to? (flashes of light / colors / patterns etc.) I notice that I am most triggered by flashes of light, both in real life and online, for example in movies. Certain tones in music can also make me feel not well. Furthermore, a quick change of black and white is also "triggering" for me.

Do you feel a seizure coming when you are confronted with a trigger and is it by then too late to look away or do something about it? I do not immediately feel a seizure, but I notice clearly that I do not feel well when I am being exposed to triggers. I do know that at that moment, I have to move away from the triggers, because there is a good chance that a seizure will follow. In addition, I also have attacks that are not visible, fortunately I hardly notice those myself.

Do you feel there are many aids or remedies for people with photosensitive epilepsy? Nowadays, there are enough medicines for epilepsy in general, for photosensitive epilepsy there are almost no special medicines because the regular medicines against epilepsy work well enough. I myself use a different type of medication, so that I do not suffer from weight gain. Many medicines for people who suffer from epilepsy do have this side effect.

Do you think you should be more warned about online triggers (movies, sites, games, etc.)? In my opinion, there is indeed a need to warn more about triggers. In real life this is of course difficult, however something could easily be made online to indicate in advance that a website or video contains a trigger. I would like to have such a warning for movies and games, but also on social media platforms such as instagram and facebook. They do have a warning on Instagram for when a video contains "intense content", so I wonder why this has not yet been done for incentives for epilepsy.

B Flowchart

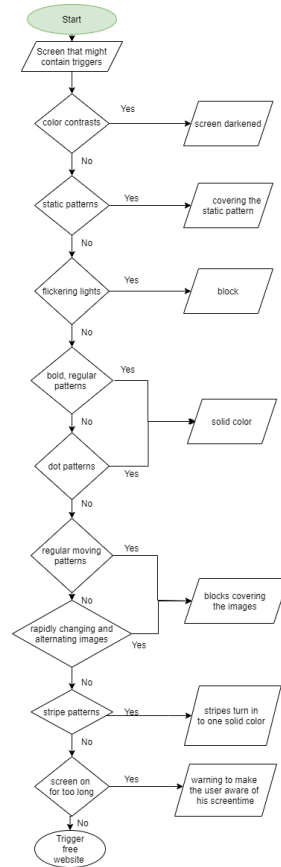


Fig. 1. Flowchart

C Power analysis

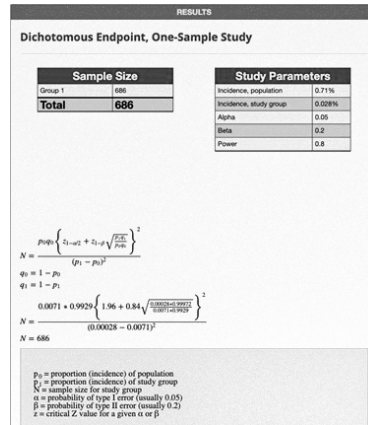


Fig. 2. Power analysis